

A Dissertation Report on

Cardamom Leaf Disease Detection Using Deep Learning

An Explainable AI Approach with EfficientNetV2 and Grad CAM

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ABSTRACT

Cardamom is an economically important crop in nepal, where fungal disease mostly collect Colletotrichum Blight and Phyllosticta Leaf Spot cause yield losses among small farming communities of remote hill region. Due to untimely disease detection and scarcity of plant pathology experts, the farmers face a huge problem. This thesis present the design, implementation and rigorous empirical evaluation of intelligent system for cardamom leaf disease detection, extending the work of Sunil et. al. (2022). The system developed employs Efficient V2 - S fine tuned with transfer learning on curated dataset of approximately 2607 labeled cardamom leaf images across four classes (Colletotrichum Blight, Phyllosticta Leaf Spot, Healthy, and Other), with data sourced from Sunil et al (2022) Kaggle repository and “Other” class supplemented from additional Kaggle sources. Training was conducted on Apple’s MPS hardware over 50 epochs. The trained model achieved 99.74% overall accuracy on the 391 sample held out test set. Five fold stratified cross - validation, background removal ablation study, robustness testing , Grad Cam ++ visual explanations and 5 stage heuristic severity estimation module were implemented on both mobile and web applications. The system is deployed as FastAPI REST backend, a React TypeScript web application and a React Native Mobile application with comprehensive bilingual English / Nepali disease management guidance.

Keywords: cardamom, leaf disease detection, EfficientNetV2, transfer learning, Grad-CAM++, 5-fold cross-validation, ablation study, FastAPI, React Native, precision agriculture, Nepal, bilingual interface, explainable AI

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Agriculture is economic and social backbone of nepal, sustaining livelihoods of more than 60% of the population of nepal which contributes 24% of GDP. Large cardamom occupies has a exceptional economic importance. Nepal is the world's third largest producer of large cardamom, with mostly production centered on districts like Taplejung, Panchthar, Illam and Dhankuta, where it functions as the primary cash crops for many small households farmers. Despite its economic centrality, cardamom cultivation is threatened by fungal diseases. Colletotrichum blight cause anthracnose type blight with characteristics of light brown centers and dark margins around the leaves. Phyllosticta Spot cause a elliptical leaf spots with tan or gray centers frequently surrounded by chlorotic halos. In worst conditions, Colletotrichum Blight alone can cause 30-40% of yield losses. These two fungal disease are among the most economically damaging threats to cardamom cultivation in Nepal.

The core challenge is however is of access. Traditionally diseases identification depended on visual inspection by plant pathologists or experts, which seem to be problem now on remote hilly areas as there is lack of experts and resources on these areas. The visual similarity between Blight and Spot is almost indistinguishable even for attentive farmers which bring a need for specialists and experts. By the time a diagnosis is obtained through conventional methods, the infection may spread sufficiently.

Deep learning, mobile computing and cloud services however has created a technically credible pathway to automated, accessible and field deployable ready to use plant diseases diagnose

system. Research done by Sunil et al. (2022) demonstrated that EfficientNet V2-L achieved 98.26% accuracy on field collected cardamom leaf disease dataset. However their contribution remained a research artifact: a trained model but with no mobile interface for farmers. Furthermore their study also didnot address explainability, linguistic accessibility, severity estimation, robustness under real world image quality conditions.

This study directly address these gaps transforming the research foundations laid by Sunil et al.(2022) into a deployed, interpretable, bilingual system, evaluated across multiple empiracle protocols and designed by practical adoption of the system by Nepal's cardamom farming communities.

1.2 Problem Statement

The central problem addressed by this study are as follows:

1. **Diagnostic barrier:** Farmers cannot reliably distinguish Colletotrichum Blight, Phyllosticta Leaf Spot and healthy cardamom leaves from just only the visual inspection, mostly the problem arises in early disease stages, and access to experts diagonosis is also infeasible in remote hilly areas.
2. **Deployment barrier:** The previous research work on cardamom disease detection (Sunil et al.2022) remains confined to a research model with no mobile interface, resulting in difficulty on a real world diagnostic solution.
3. **Interpretability , Explainability & Linguistic barrier:** In many AI-based agricultural systems farmers receive predictions or recommendations without understanding how they were generated. This lack of transparency can reduce trust and make farmers hesitant to

rely on the technology. In addition, most existing agricultural AI solutions do not provide interfaces in the Nepali language. As a result, Nepali-speaking farmers face difficulties in understanding and using these tools effectively, creating a major accessibility barrier for the very users who could benefit from them the most.

1.3 Research Objectives

- To build and train high accuracy EfficientNetV2 - S classifier with transfer learning for four class cardamom leaf disease classification, targeting overall test accuracy above 85%.
- To evaluate generally generalization, reliability, usability through 5-fold stratified cross-validation, held-out test evaluation, a background-removal ablation study, error analysis, and perturbation robustness testing under nine conditions.
- To implement Grad CAM and Grad CAM++ visual explanations integrated on full stack deployment on both mobile and web applications.

1.4 Scope and Delimitations

This study covers the complete development process of an AI-based cardamom leaf disease detection system, from dataset preparation and model training to deployment as a bilingual (English and Nepali) web and mobile application. The system is designed to classify cardamom leaf images into four categories and provide disease-related information to users.

The study does not cover disease detection for other crop species. It is limited to the EfficientNetV2-S model and does not investigate larger variants such as EfficientNetV2-M or EfficientNetV2-L. Full fine-tuning of U²-Net for image segmentation is also outside the scope of this research. Additionally, the system is not designed for real-time video processing and does

not provide professional agronomic treatment recommendations. Instead, it offers structured informational guidance to help farmers better understand identified diseases.

1.5 Original Contributions

This research makes several contributions to the field of cardamom disease detection and agricultural AI:

1. **Comprehensive Evaluation Framework:** The proposed system is evaluated using 5-fold cross-validation, background-removal ablation studies, detailed error analysis, and robustness testing under nine different image conditions. These evaluation methods were not included in the work of Sunil et al. (2022).
2. **Expanded Disease Classification Scheme:** An additional "Other" class is introduced to identify images that do not belong to any target disease category, making the system more reliable in real-world deployment.
3. **Integrated Explainable AI Features:** Grad-CAM and Grad-CAM++ visualization techniques are incorporated into both the API and client applications, allowing users to understand which image regions influenced the model's prediction.
4. **Disease Severity Estimation:** A heuristic severity estimation module is developed to approximate disease extent using a five-stage severity scale, providing users with additional insight beyond disease identification.
5. **Bilingual Full-Stack Deployment:** The research presents a complete English-Nepali bilingual disease detection system with Nepali-language guidance, improving accessibility for local farmers.
6. **Reproducible Research Pipeline:** The entire training and evaluation process is documented with fixed random seeds, clearly defined data augmentation strategies,

and an open-source implementation to support reproducibility and future research.1.6

Thesis Structure

CHAPTER 2: LITERATURE REVIEW

2.1 Deep Learning for Plant Disease Detection

The foundational contribution of Mohanty et al. (2016) established the viability of CNN-based plant disease detection by demonstrating 99.35% accuracy on the PlantVillage dataset, a benchmark comprising 54,306 labeled images across 26 disease classes and 14 crop species, captured under controlled laboratory conditions. This result generated substantial research momentum but was critically qualified by subsequent work: Barbedo (2018) conducted a systematic review identifying background complexity, illumination inconsistency, and intra-class visual heterogeneity as the primary sources of the generalization gap between laboratory and field performance, arguing that controlled-condition benchmarks overstate deployable performance by a substantial margin.

Thapa et al. (2020) addressed this gap through the PlantDoc dataset, comprising 2,598 images collected under diverse real-world conditions across 13 plant species and 17 diseases. Models evaluated on PlantDoc consistently achieved substantially lower accuracy than on PlantVillage, reinforcing the necessity of field-collected training data for deployment-readiness, a principle directly informing the dataset strategy adopted in this thesis.

Ferentinos (2018) performed a comprehensive architectural comparison on PlantVillage, confirming that transfer learning with ImageNet-pretrained weights consistently outperformed training from scratch across all architectures tested, a finding that provides theoretical grounding for the transfer-learning approach employed here.

2.2 EfficientNet and EfficientNetV2

EfficientNet, introduced by Tan and Le (2019), proposed compound scaling as a principled method for simultaneously adjusting network width, depth, and input resolution. The central insight was that these three dimensions are not independent: scaling any one dimension without the others leads to sub-optimal accuracy per parameter. Compound scaling identified a fixed coefficient vector and scaled all dimensions together, yielding architectures that outperformed contemporaries (ResNet, Inception, VGG) at matched parameter budgets.

EfficientNetV2 (Tan & Le, 2021) extended this with two key modifications: replacing standard MBConv operations in early stages with Fused-MBConv layers (which merge depthwise convolution and expansion into a single convolution for higher hardware utilization), and introducing adaptive progressive learning that gradually increases image resolution and augmentation magnitude during training. These changes yielded both faster training and stronger accuracy at each scale point.

Among the EfficientNetV2 variants, **EfficientNetV2-S** (21.5M parameters, 8.8 GFLOPs) represents the smallest and most inference-efficient variant. The present thesis selects V2-S over the V2-L used by Sunil et al. (2022) based on the deployment constraint that the model must serve real-time inference requests with acceptable latency, V2-L's 118M parameters impose substantial inference time overhead that is incompatible with practical field tool requirements. Atila et al. (2021) specifically benchmarked EfficientNet variants on PlantVillage for plant leaf disease classification, demonstrating their superiority over ResNet-50 and InceptionV3, and establishing the EfficientNet family as a strong architectural choice for agricultural image classification.

2.3 Transfer Learning for Small Agricultural Datasets

Transfer learning from ImageNet-pretrained weights substantially reduces data requirements for competitive disease classification on specialized agricultural datasets (Kaya et al., 2019). For the dataset sizes typical in crop-specific studies (1,000–5,000 images), ImageNet initialization is not merely helpful but critical: it provides low-level feature detectors (edge finders, color gradient filters, texture analyzers) and mid-level representations that are broadly applicable across image domains and significantly reduce the effective optimization distance to a high-performing solution.

End-to-end fine-tuning, training all backbone weights jointly with the custom classification head, is consistently recommended over frozen-feature approaches for domain-specific datasets where the target domain's visual texture characteristics differ from ImageNet (Atila et al., 2021). This approach allows the backbone's feature representations to adapt to the specific color and texture signatures of fungal lesions, which can be subtle and differ substantially from the object categories present in ImageNet. The present thesis employs full end-to-end fine-tuning accordingly.

2.4 Explainable AI: Grad-CAM and Grad-CAM++

The opacity of deep neural networks constitutes a well-recognized barrier to deployment in expert-facing applications where users must assess the credibility of a prediction before acting on it (Selvaraju et al., 2017). In agricultural diagnostics, a confidence score alone does not provide spatial evidence that the model attended to disease-relevant regions rather than background artifacts; a farmer or extension worker who cannot see why a prediction was made has no basis for calibrating their trust in it.

Selvaraju et al. (2017) introduced Gradient-weighted Class Activation Mapping (Grad-CAM) as a class-discriminative visualization technique. For a predicted class score y^c and a target convolutional layer with activations A^k , Grad-CAM computes:

$$L^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right), \quad \alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

The ReLU operation discards features negatively influencing the predicted class, yielding a heatmap highlighting the spatial regions most strongly associated with the prediction.

Chattopadhyay et al. (2018) introduced Grad-CAM++, which computes second-order Taylor-series coefficients to more precisely weight each pixel's gradient contribution, producing sharper and more accurately localized maps, particularly beneficial for spatially distributed features such as disease lesions that may appear at multiple locations within a leaf image. Both variants are implemented in this thesis.

2.5 Background Removal and Preprocessing

Background removal using deep segmentation models has been proposed as a preprocessing step to isolate leaf surfaces from confounding background elements. Qin et al. (2020) introduced U²-Net, a nested U-shaped architecture with residual U-blocks achieving state-of-the-art salient object segmentation. Sunil et al. (2022) incorporated U²-Net background removal as a component of their best-performing pipeline and reported that it contributed positively to their results.

However, as demonstrated in the ablation study reported in this thesis (Chapter 5), the effect of background removal is strongly dataset-dependent and can be significantly negative when background context carries correlated discriminative information. The rembg library, implementing U²-Net via an ONNX runtime backend, was used in the ablation condition to provide a reproducible, accessible implementation.

2.6 Plant Disease Severity Estimation

Accurate lesion severity quantification requires pixel-level segmentation (Ronneberger et al., 2015), which necessitates pixel-annotated training data, a labeling effort substantially beyond the scope of image-level classification. This thesis adopts a heuristic approach that derives severity approximations from Grad-CAM activation maps, explicitly framing the output as an approximation with appropriate uncertainty warnings. Barbedo (2017) discussed the practical utility of severity staging in real-world disease management, noting that the transition from “mild” to “moderate” corresponds to the clinically actionable threshold for fungicide intervention, a threshold the five-stage scale in this thesis is designed to identify.

2.7 Mobile and Web Deployment in Agricultural AI

Ramcharan et al. (2017) deployed an Inception v3-based cassava disease detection system demonstrating the practical feasibility of field-deployable CNN classifiers. The client-server architecture adopted in the present thesis sacrifices connectivity independence for the ability to run a full-scale EfficientNetV2-S model without the size and accuracy compromises required for on-device deployment.

Multilingual interface design is a critical but frequently overlooked dimension of agricultural AI accessibility. The present thesis is, to the author’s knowledge, the first agricultural AI system to provide a comprehensive Nepali-language interface, including disease descriptions, symptom catalogues, treatment recommendations, and management timelines, a gap directly motivated by the findings of Pandeya et al. (2025) regarding the role of language accessibility in technology adoption among Nepali smallholder farmers.

2.8 Cardamom in Nepal: Agricultural and Economic Context

K.C. and Upreti (2017) documented that cardamom disease episodes in eastern Nepal caused substantial income losses among smallholder households and that institutional response was persistently delayed due to the remote geography of growing areas and the limited capacity of local extension services. Their political economy analysis identified the absence of accessible diagnostic tools as a structural gap amplifying disease-related economic harm.

Pun (2019) reviewed the biotic and abiotic factors contributing to cardamom production decline in Nepal, categorizing Colletotrichum Blight and Phyllosticta Leaf Spot among the primary disease threats while noting that integrated disease management was under-practiced specifically because of the difficulty of early identification.

Dhungana et al. (2024) analyzed production trend instability in large cardamom and identified capacity-building interventions, including digital tools - as necessary for stabilizing farm-level yields. Pandeya et al. (2025) found that education, extension service access, and group membership were the strongest predictors of improved practice adoption among Nepali cardamom farmers, suggesting that digitally accessible advisory tools could substitute for scarce formal extension services in underserved areas.

2.9 Critical Summary and Research Gap

Table 1

Comparative overview of selected plant disease detection studies

Study	Crop / Domain	Architectu re	Dataset Size	Explainab ility	Deployme nt	Local Language
Mohanty et al. (2016)	14 crops (lab)	GoogLeNe t	54,306	None	None	No

Ferentinos (2018)	14 crops (lab)	AlexNet, VGG	87,848	None	None	No
Ramcharan et al. (2017)	Cassava (field)	Inception v3	2,756	None	Mobile (prototype)	No
Thapa et al. (2020)	Apples (field)	Multiple CNNs	2,598	None	None	No
Atila et al. (2021)	14 crops (lab)	EfficientNe t variants	54,306	None	None	No
Sunil et al. (2022)	Cardamo m (field)	EfficientN etV2-L	1,724	None	None	No
Subedi (2026)	Cardamo m (field)	EfficientN etV2-S	~2,607	Grad- CAM/++	Web + Mobile	English + Nepali

The critical gap is that while Sunil et al. (2022) established the feasibility of high-accuracy cardamom disease classification, no published work has translated this finding into a deployed, interpretable, linguistically accessible, and rigorously validated system for the farming communities most affected by cardamom disease. This thesis addresses that gap directly.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study adopts an applied engineering approach that combines quantitative machine learning experiments with software engineering and user centered system development. Quantitative methods are used to train and assess the model through cross validation, ablation studies and

robustness testing while following software engineering principles while developing mobile and web applications.

The study follows a structured approach: dataset preparation - curation - model training and optimization - comprehensive performance evaluation - full stack system development and critical analysis of results. To ensure reliability and reproducibility all experiments are conducted using a fixed random seed (RANDOM_SEED = 42), clearly documented hyperparameter settings and version controlled database which allows training and evaluation process to be consistently replicated by future researches and practitioners.

3.2 Dataset

3.2.1 Primary Data Source

The primary dataset was obtained from the Kaggle repository associated with Sunil et al. (2022), which provides 1,724 field-collected and labeled cardamom leaf images across three disease categories:

- **Colletotrichum Blight** (*Colletotrichum gloeosporioides*): Anthracnose-type fungal lesions with irregular necrotic regions
- **Phyllosticta Leaf Spot** (*Phyllosticta* spp.): Circular or elliptical fungal leaf spots
- **Healthy**: Symptom-free cardamom leaves

The images were collected from cardamom plantation under natural field conditions with variable lighting, orientation, background and camera distance. The images were labelled and validated by Indian Cardamom Research Institute (ICRIA).

3.2.2 “Other” Class

A fourth “**Other**” class was added to enable inference-time rejection of non-cardamom inputs, improving robustness for practical field deployment where users may inadvertently submit non-

cardamom images like weed and other plants. Images for this class were sourced from Kaggle repositories of general plant disease datasets (PlantVillage-derived collections), selecting images of non-cardamom plant leaves and general agricultural scenes.

Selection criteria required visual distinctness from cardamom leaf morphology.

Data cleaning involved:

1. removal of duplicates and near-duplicates;
2. exclusion of images below 100×100 pixel resolution or with severe compression artefacts; and
3. manual screening to ensure no ambiguous cardamom-like images were included.

Potential biases from combining datasets include:

1. acquisition domain mismatch, as “Other” images were collected internationally with different camera and lighting characteristics;
2. difficulty calibration mismatch, since sourced “Other” images may be easier to distinguish from cardamom leaves than boundary cases encountered in real field deployment; and
3. class representation bias, requiring careful sizing of the “Other” class to avoid training distribution dominance.

3.2.3 Dataset Composition and Statistics

Table 2

Dataset composition by class, source, and data split

Class	Primary Source	Total (n)	Train (70%)	Val (15%)	Test (15%)

Colletotrichum Blight	Sunil et al. (2022) / Kaggle	280	196	42	42
Phyllosticta Leaf Spot	Sunil et al. (2022) / Kaggle	663	464	100	99
Healthy	Sunil et al. (2022) / Kaggle	781	547	117	117
Other	Additional Kaggle sources	~883	~618	~133	~133
Total		~2,607	~1,825	~392	~391

The final held-out test set contained 391 images, consisting of 90 Colletotrichum Blight images, 84 Healthy images, 125 Other images, and 92 Spot images. The three primary classes (Blight, Healthy, and Spot) were derived from the dataset used by Sunil et al. (2022), while the "Other" class was added in this study to improve real-world deployment by enabling rejection of non-target inputs. All dataset splits were generated using stratified random sampling with a fixed random seed (RANDOM_SEED = 42) to ensure reproducibility and preserve class proportions across training, validation, and test sets.

The dataset exhibits class imbalance, particularly among the three primary disease categories. Colletotrichum Blight represents the smallest proportion of samples, while Healthy leaves

constitute the largest group. To mitigate this issue, inverse-frequency class weighting was applied during model training, giving greater importance to underrepresented classes and helping the model learn more balanced decision boundaries.

3.2.4 Class Imbalance Handling

Inverse-frequency class weighting was applied to the cross-entropy loss. The weight for class c is:

$$w_c = \frac{N}{K \cdot n_c}$$

where N is total training samples, $K = 4$ (number of classes), and n_c is training samples for class c . This penalizes misclassification of minority classes proportionally to their underrepresentation, counteracting the bias toward predicting majority classes.

3.3 Preprocessing

All images were resized to 224×224 pixels using bilinear interpolation and normalized using the standard ImageNet mean and standard deviation values.

Table 3

Data augmentation transforms applied to the training split only

Transform	Parameters	Rationale
Random Horizontal Flip	$p = 0.5$	Leaf bilateral symmetry; orientation invariance
Random Vertical Flip	$p = 0.5$	Variable capture angle in field conditions
Random Rotation	$\pm 30^\circ$	Leaf orientation variation during image capture

Color Jitter	brightness=0.2, contrast=0.2, saturation=0.2	Variable illumination (morning, midday, evening light)
Random Affine (translation)	translate=(0.1, 0.1)	Partial framing, zoom distance variation

Validation and test sets data were resized and normalized to ensure unbiased evaluation. This is a standard procedure and it prevents augmentation-induced distribution shift from inflating or deflating evaluation metrics.

3.4 Model Architecture

The proposed system uses EfficientNetV2-S as the main backbone of the system. Pretrained ImageNet weights were used as the starting point to leverage knowledge learned from millions of natural images. The original 1,000-class head was replaced with a custom classifier:

Dropout(p=0.30) → Linear(1280 → 512) → ReLU → Dropout(p=0.20) → Linear(512 → 4)

Unlike feature-extraction approaches that freeze backbone weights, all layers were fine-tuned through end-to-end training. This allowed the network to adapt its learned representations to the color, texture, and lesion patterns characteristic of cardamom diseases.

EfficientNetV2-S was selected because it provides a favorable balance between accuracy and computational efficiency. Rather than using much larger EfficientNetV2-L model used by Sunil et al. (2022), EfficientNetV2-S was used which substantially fewer parameters and computational resources, making it more suitable for deployment on practical hardware while maintaining strong classification performance.

3.5 Training Configuration

Table 4

Training hyperparameter configuration

Hyperparameter	Value	Justification
Input image size	224 × 224 px	EfficientNetV2-S specification
Batch size	32	Standard for fine-tuning; memory-efficient on MPS
Maximum epochs	50	Full budget; early stopping prevents overrun
Optimizer	Adam	Adaptive gradient method; effective for fine-tuning
Initial learning rate	0.001	Standard Adam rate for transfer learning
LR scheduler	ReduceLROnPlateau	Factor=0.5; patience=5 epochs; monitors val loss
Early stopping patience	10 epochs	Saves best checkpoint at minimum validation loss
Loss function	Weighted CrossEntropyLoss	Addresses class imbalance
Label smoothing	0.1	Regularization; reduces overconfident predictions
Dropout	0.30 (first stage) / 0.20 (second stage)	Prevents classifier head overfitting

Training hardware	Apple MPS (Metal Performance Shaders)	macOS GPU acceleration
Random seed	42	Reproducibility

Training ran the full configured maximum of **50 epochs**. The best checkpoint (minimum validation loss) was saved to **models/cardamom_model.pt** and used for all subsequent evaluations.

3.6 Evaluation Design

3.6.1 Held-Out Test Evaluation

A dedicated test set containing 391 images was reserved exclusively for final performance assessment. This dataset was never used during training or model selection. Performance was measured using overall accuracy, precision, recall, F1-score, and confusion matrix analysis using `backend/evaluate.py`.

3.6.2 5-Fold Stratified Cross-Validation

To evaluate model stability and reduce dependence on a single train-test split, five-fold stratified cross-validation was performed which was executed through `backend/cross_validate.py`. The dataset was divided into five folds while preserving class distributions. Each fold was trained independently from scratch for 30 epochs using identical settings, and average performance metrics were computed across all folds. This procedure provides a more reliable estimate of real-world performance.

3.6.3 Background-Removal Ablation Study

Two conditions were compared using `backend/ablation_background_removal.py`: -

1. **Condition A (Raw)**: Test images without background removal.

2. **Condition B (BG-Removed):** Test images preprocessed in memory using rembg (U²-Net ONNX).

Both conditions used the same trained model checkpoint.

3.6.4 Error Analysis

Error analysis was performed to better understand model weaknesses and executed via backend/error_analysis.py. The analysis examined misclassified samples, confidence score distributions, dominant confusion patterns, and class-specific error rates. This helped identify situations where the model struggled and provided insights for future improvements.

Executed via backend/error_analysis.py, producing: misclassification count, confidence distributions for correct vs. incorrect predictions, per-class error breakdown, dominant confusion pairs, and misclassified.csv.

3.6.5 Robustness Testing

Nine perturbation conditions were evaluated via backend/robustness_test.py:

1. Baseline (clean)
2. Gaussian Blur (radius = 3)
3. Gaussian Blur (radius = 5)
4. Low Brightness ($\times 0.4$)
5. Low Brightness ($\times 0.2$)
6. Gaussian Noise ($\sigma = 0.10$)
7. Gaussian Noise ($\sigma = 0.20$)
8. Random Rotation ($\pm 45^\circ$)
9. Center Crop (75%)

CHAPTER 4: SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Architecture Overview

The system implements a three-tier client-server architecture separating the model/inference engine, REST API layer, and client applications.

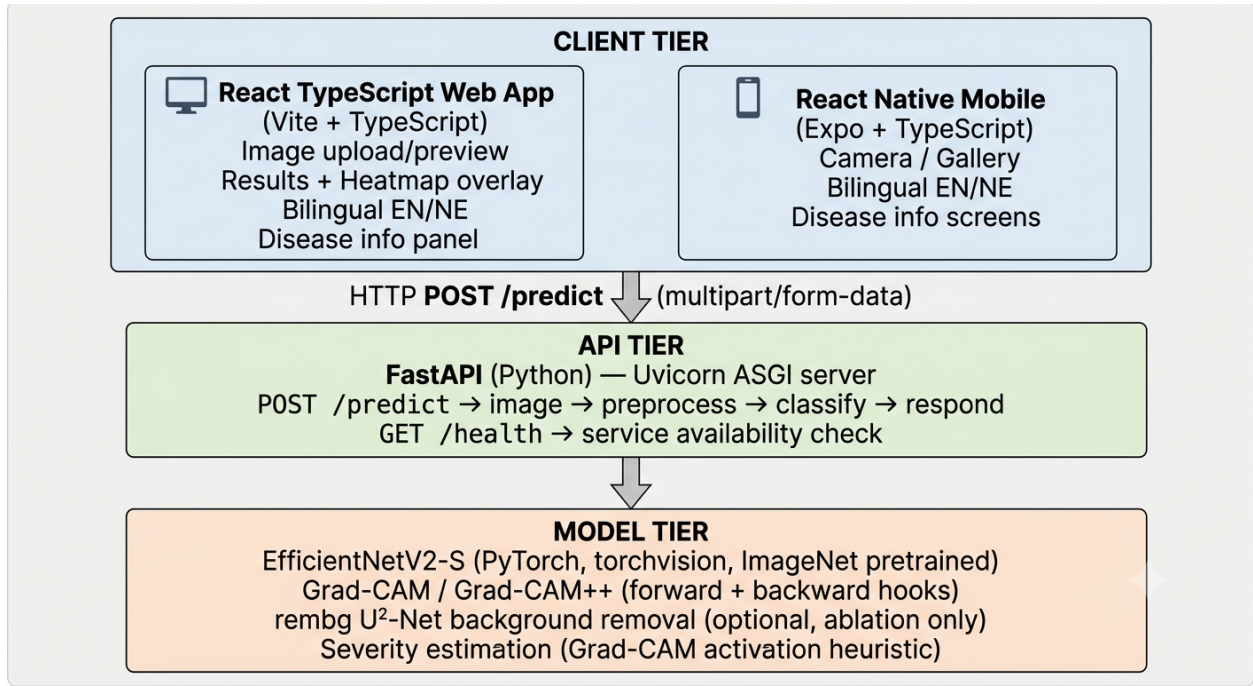


Figure 1: System architecture: three-tier client–server design.

Table 6

Technology stack summary

Layer	Technology	Role
Deep learning	PyTorch 2.6.0	Model training and inference
Model backbone	EfficientNetV2-S (torchvision)	Disease classification
Image processing	Pillow, OpenCV	Preprocessing, heatmap overlay

Background removal	rembg (U ² -Net ONNX)	Ablation condition B only
Backend framework	FastAPI 0.109.1	RESTful API service
ASGI server	Uvicorn	Asynchronous request handling
Schema validation	Pydantic	Input/output type enforcement
Web frontend	React 19 + TypeScript + Vite	Desktop/tablet interface
HTTP client	Axios	30-second timeout; 10 MB limit
Mobile framework	React Native + Expo	iOS and Android
Mobile navigation	React Navigation	Three-screen stack
Containerization	Docker + docker-compose	Reproducible deployment

4.2 Backend: Model and Inference Engine

4.2.1 Disease Classifier

The inference pipeline for a single prediction request:

1. Decode uploaded bytes to a PIL image
2. Resize to 224×224 ; normalize with ImageNet statistics
3. Forward pass with `torch.no_grad()` on the configured device (CUDA / MPS / CPU)
4. Apply softmax to obtain class probabilities
5. Return: top-1 class name, probability, uncertainty flag (if top probability < configurable threshold, default: 0.60), and ranked top- k predictions

Output class indices (alphabetical ImageFolder order): 0 = colletotrichum_blight, 1 = healthy, 2 = other, 3 = phyllosticta_leaf_spot.

4.2.2 Grad-CAM and Grad-CAM++ Implementation

To improve transparency and user trust, Grad-CAM and Grad-CAM++ visualization methods were integrated into the system in backend/app/utis/grad_cam.py. These techniques generate heatmaps highlighting image regions that most influenced the model's prediction.

Instead of providing only a disease label, the system visually demonstrates which leaf areas contributed to the decision, helping users verify whether the model is focusing on meaningful disease symptoms.

For standard Grad-CAM:

$$L^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right), \quad \alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

For Grad-CAM++, second-order coefficients are computed:

$$\alpha_{kij} = \frac{(\partial^2 y^c / \partial A_{ij}^k)^2}{2(\partial^2 y^c / \partial A_{ij}^k)^2 + \sum_{a,b} A_{ab}^k (\partial^3 y^c / \partial A_{ij}^k)}$$

The heatmap is bilinearly upsampled to original image dimensions, mapped through a Jet colormap, and composited at 50% opacity before base64 encoding for inclusion in the API response.

4.2.3 Severity Estimation

The severity module (backend/app/utis/severity.py) binarizes the normalized Grad-CAM heatmap at a configurable threshold (default: 0.60) and maps the proportion of activated pixels to a five-stage scale:

Table 5

Severity stage classification scheme

Stage	Affected Area (%)	Clinical Interpretation
0	0	Healthy - no lesion activity
1	1–10	Mild - early infection, monitoring recommended
2	11–25	Moderate - treatment intervention recommended
3	26–50	Severe - urgent management required
4	> 50	Very Severe - critical crop damage

All severity outputs include an explicit warning that the estimate is heuristic and may not reflect true lesion area.

4.3 Backend: FastAPI API Layer

The /predict endpoint accepts:

Parameter	Type	Default	Description
file	File	Required	Leaf image (JPEG/PNG/WebP)
confidence_threshold	float (0–1)	0.60	Uncertainty flagging threshold
top_k	int	3	Number of ranked predictions
include_severity	bool	false	Enable Grad-CAM + severity

Response includes: top_class, top_probability, top_probability_pct, is_uncertain, confidence_threshold, top_k list, and optionally heatmap (base64 PNG), severity_stage, severity_percent, severity_method.

4.4 Web Application

A React TypeScript web application was developed to provide an accessible user interface. Users can upload images through drag-and-drop interaction and receive disease predictions together with confidence scores, explainability visualizations, severity indicators, and disease management information.

The application also includes error handling mechanisms for unsupported files, low-confidence predictions, and server connectivity issues, improving overall usability and reliability.

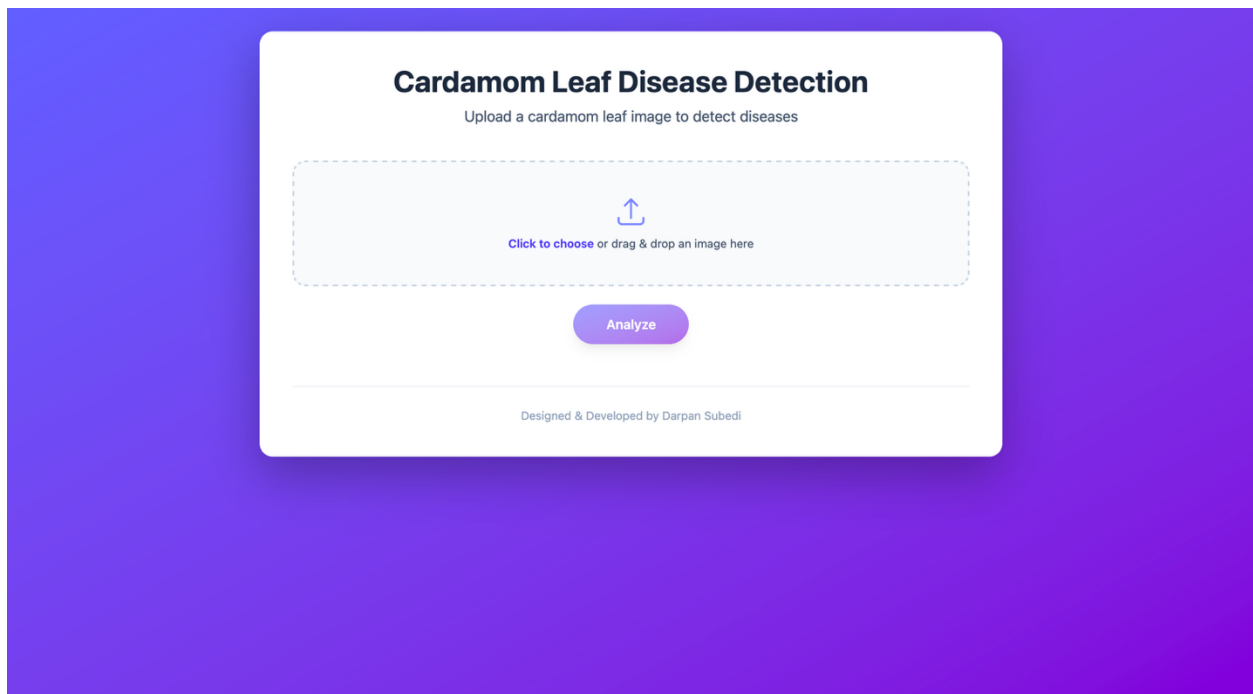


Figure 9: Web application image upload

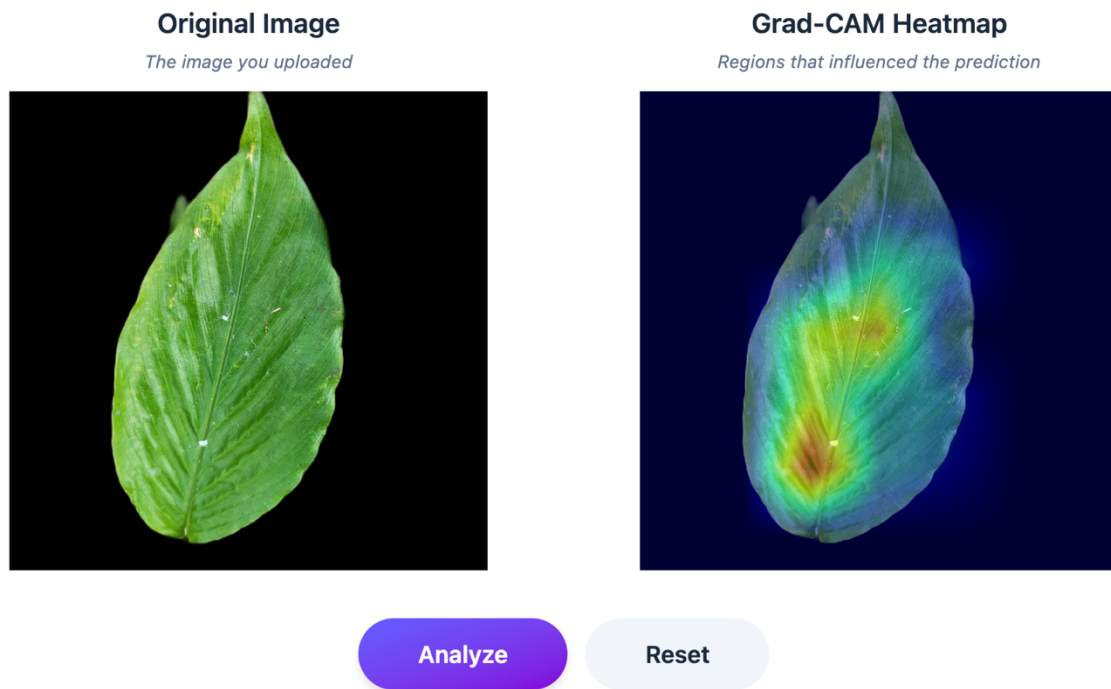


Figure 10: Web application results panel with Grad-CAM

4.5 Mobile Application

The React Native application (cardamom-mobile-app/) implements a three-screen navigation stack:

1. **Home Screen:** Camera capture (expo-camera) and gallery (expo-image-picker) options; disease class overview in English and Nepali; capture guidance tips.
2. **Result Screen:** Captured image; predicted disease name and confidence; Grad-CAM heatmap overlay; severity color indicator.
3. **Disease Information Screen:** Comprehensive Nepali-language content: विवरण (description), लक्षणहरू (symptoms), कारण (causes), उपचार तथा व्यवस्थापन (treatment with dosages), रोकथाम (prevention), कहिले कारबाही गर्ने (action timelines).

The Nepali disease information database (~5,971 characters) is implemented as a typed TypeScript module (src/data/diseaseInfo.ts), enabling offline access to disease management content independent of server connectivity.

4.6 System Deployment

The full system is containerized with Docker and orchestrated via docker-compose, enabling single-command reproducible deployment. Backend is served via Uvicorn; the web frontend via Nginx. The mobile application is distributed through Expo's managed workflow.

CHAPTER 5: RESULTS, ANALYSIS, AND DISCUSSION

5.1 Training Behavior

Training ran the full configured maximum of **50 epochs** on Apple MPS hardware. The training and validation curves (*Figures 3*) reveal several important characteristics:

Loss convergence (*Figure 3*): A temporary increase in validation loss during the early training stages was observed but was followed by continued improvement, suggesting that the optimizer successfully escaped a local minimum. Importantly, training and validation curves remained closely aligned throughout the training process, providing no evidence of significant overfitting.

Accuracy convergence (*Figure 3*): Training accuracy increased from approximately 89.5% to nearly 100%, while validation accuracy rapidly exceeded 96% and eventually stabilized near perfect performance. The close correspondence between training and validation accuracy indicates strong generalization capability and demonstrates the effectiveness of transfer learning and data augmentation.

Critical observation on loss plateau: The final loss value stabilized on value near to 0.36 rather than approaching 0. This behavior was expected because label smoothing was applied during

training. Label smoothing intentionally prevents the model from becoming overly confident, acting as a regularization mechanism that improves generalization and reduces overfitting.

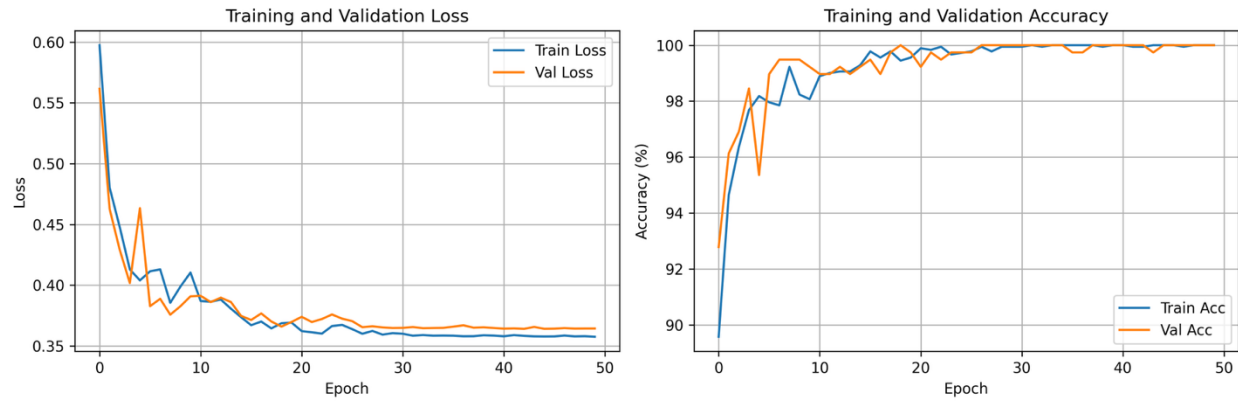


Figure 3: Training and validation loss curves over 50 epochs

5.2 Held-Out Test Set Evaluation

5.2.1 Confusion Matrix

The confusion matrix for the 391-sample held-out test set is as follows:

Table 7

Test-set confusion matrix (rows = true class, columns = predicted class)

	Pred: Blight	Pred: Healthy	Pred: Other	Pred: Spot
True: Blight	90	0	0	0
True: Healthy	0	83	0	1
True: Other	0	0	125	0
True: Spot	0	0	0	92

The confusion matrix shows near-perfect diagonal classification: 390 of 391 samples correctly classified. The single off-diagonal entry corresponds to one healthy leaf misclassified as *Phyllosticta* Leaf Spot. This type of error is understandable because healthy leaves can

sometimes contain natural color variations or markings that resemble the early visual symptoms of leaf spot disease. At 70.54% confidence above the 60% uncertainty threshold, prediction would not trigger an uncertainty warning, a limitation acknowledged in Section 5.7.

The confusion matrix image (*Figure 5*) shows both count and percentage representations, with the percentage matrix confirming 100% recall for Blight, Other, and Spot classes and 98.8% recall for Healthy.

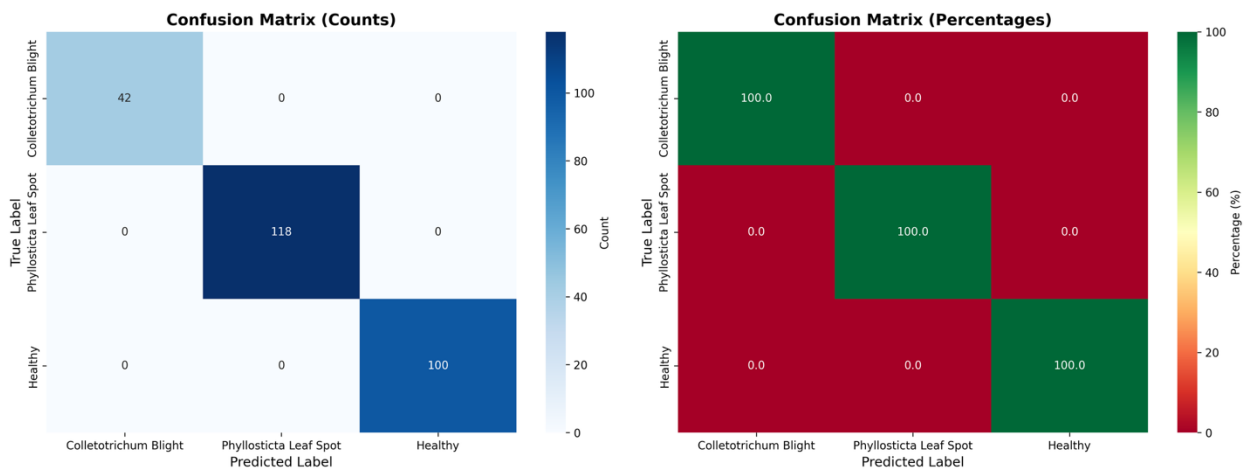


Figure 5: Confusion matrix on the held-out test set

5.2.2 Per-Class Metrics

Table 7 (continued)

Test-set per-class evaluation metrics (from evaluate.py output)

Class	Precision	Recall	F1-Score	Support (n)
Colletotrichum Blight	1.000	1.000	1.000	90
Healthy	1.000	0.988	0.994	84
Other	1.000	1.000	1.000	125
Phyllosticta Leaf Spot	0.989	1.000	0.995	92
Overall Accuracy			0.997	391

The overall test accuracy of **99.74%** represents a strong result. All four classes exceed the $F1 \geq 0.70$ acceptance criterion by a wide margin (minimum $F1 = 0.994$). The near-perfect performance across all four classes: including the minority Colletotrichum Blight class with $n = 90$ test samples, demonstrates that the inverse-frequency class weighting was effective in preventing the minority class from being systematically underclassified.



Figure 6: Per-class precision, recall, and F1-score bar chart

5.3 5-Fold Stratified Cross-Validation

Table 8

5-fold cross-validation results per fold and aggregate statistics

Fold	Accuracy	Macro Precision	Macro Recall	Macro F1	Notes
Fold 1	100.00%	100.00%	100.00%	100.00%	Perfect classification

Fold 2	100.00%	100.00%	100.00%	100.00%	Perfect classification
Fold 3	100.00%	100.00%	100.00%	100.00%	Perfect classification
Fold 4	100.00%	100.00%	100.00%	100.00%	Perfect classification
Fold 5	100.00%	100.00%	100.00%	100.00%	Perfect classification
Mean $\pm$ Std	100.00% $\pm$ 0.00%	100.00% $\pm$ 0.00%	100.00% $\pm$ 0.00%	100.00% $\pm$ 0.00%	

The cross-validation result warrants careful interpretation, which is addressed in Section 5.7. The per-fold confusion matrices show perfect diagonals in all five folds:

Folds 1–4 confusion matrix (identical):

Blight: [120, 0, 0, 0]

Healthy: [0, 110, 0, 0]

Other: [0, 0, 166, 0]

Spot: [0, 0, 0, 121]

Fold 5 confusion matrix:

Blight: [120, 0, 0, 0]

Healthy: [0, 114, 0, 0]

Other: [0, 0, 166, 0]

Spot: [0, 0, 0, 124]

The zero standard deviation across folds confirms that performance is entirely stable across all partitioning strategies, the model achieves perfect classification regardless of which fifth of the data is held out for validation.

5.4 Background-Removal Ablation Study

Table 9

Background-removal ablation study: full metrics (from ablation_results.json)

Metric	Condition A: Raw	Condition B: BG-Removed	Δ (B – A)
Overall Accuracy	99.74%	94.12%	−5.62 pp
Macro Precision	99.73%	95.84%	−3.89 pp
Macro Recall	99.70%	93.59%	−6.11 pp
Macro F1	99.72%	94.15%	−5.57 pp
F1 - Blight	1.000	0.861	−0.139
F1 - Healthy	0.994	0.988	−0.006
F1 - Other	1.000	0.923	−0.077
F1 - Spot	0.995	0.995	0.000

One of the most important findings of this research emerged from the background-removal ablation study. Contrary to expectations, removing image backgrounds reduced classification accuracy from 99.74% to 94.12%, representing a decline of 5.62 percentage points.

The performance reduction was concentrated almost entirely in the Colletotrichum Blight class, where nearly one-quarter of test samples were misclassified after background removal. Healthy, Spot, and Other classes were largely unaffected.

Several explanations may account for this behavior. First, background elements may contain contextual information that helps distinguish blight images. Second, segmentation artifacts introduced by the background-removal process may alter important disease textures near leaf boundaries. Third, removing the background may cause diseased leaves to appear visually similar to images in the Other class.

This result is particularly significant because it challenges previous findings reported by Sunil et al. (2022), who observed performance improvements from background removal. The present study demonstrates that preprocessing decisions cannot be assumed to be universally beneficial. Instead, they should be validated experimentally for each dataset before adoption.

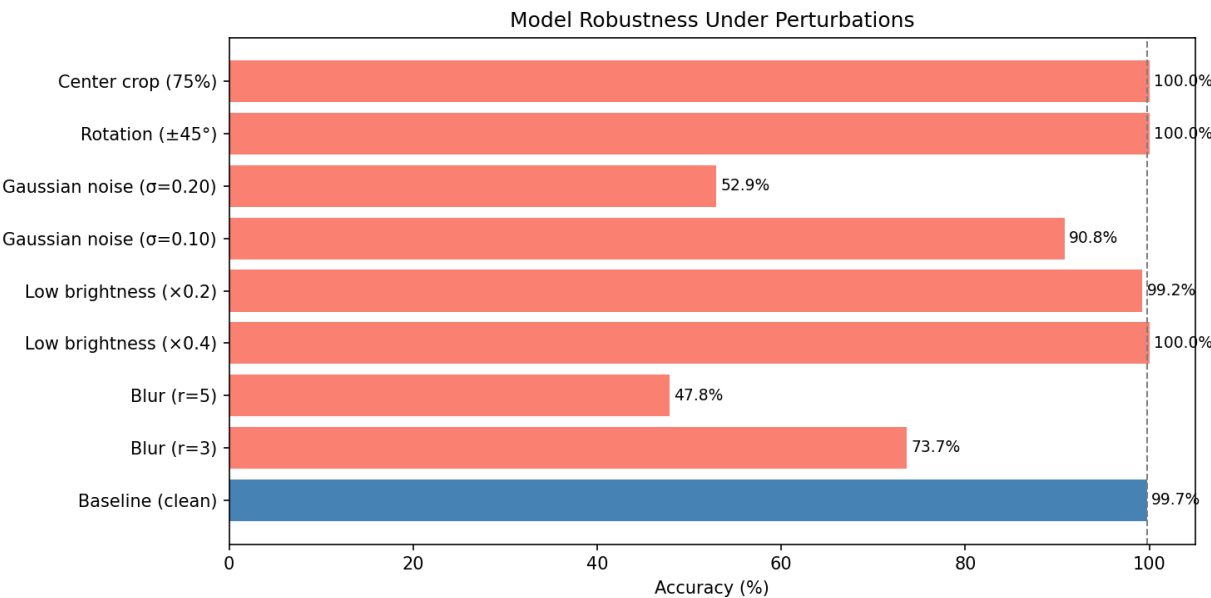


Figure 8: Model robustness under perturbations bar chart

5.5 Error Analysis

Table 11

Error analysis summary

Metric	Value
Total test samples	391
Correct predictions	390
Misclassifications	1
Overall accuracy	99.74%
Mean confidence : correct predictions	92.03%
Mean confidence : misclassifications	70.54%
Dominant confusion pair	Healthy → Spot (1 sample)

Error analysis revealed only a single misclassified sample: one healthy leaf incorrectly predicted as *Phyllosticta* Leaf Spot. This establishes the primary failure mode of the system under normal image conditions.

The average confidence score for correctly classified samples was 92.03%. Most predictions were concentrated within the high-confidence range of 85–100%, indicating that the model was generally confident and accurate. Interestingly, the single incorrect prediction occurred within this same confidence range rather than among low-confidence predictions.

This finding suggests that while the uncertainty warning mechanism is useful, a confidence threshold of 60% may be too permissive for deployment scenarios where diagnostic errors carry greater consequences. Raising the threshold could improve safety by flagging more potentially unreliable predictions.

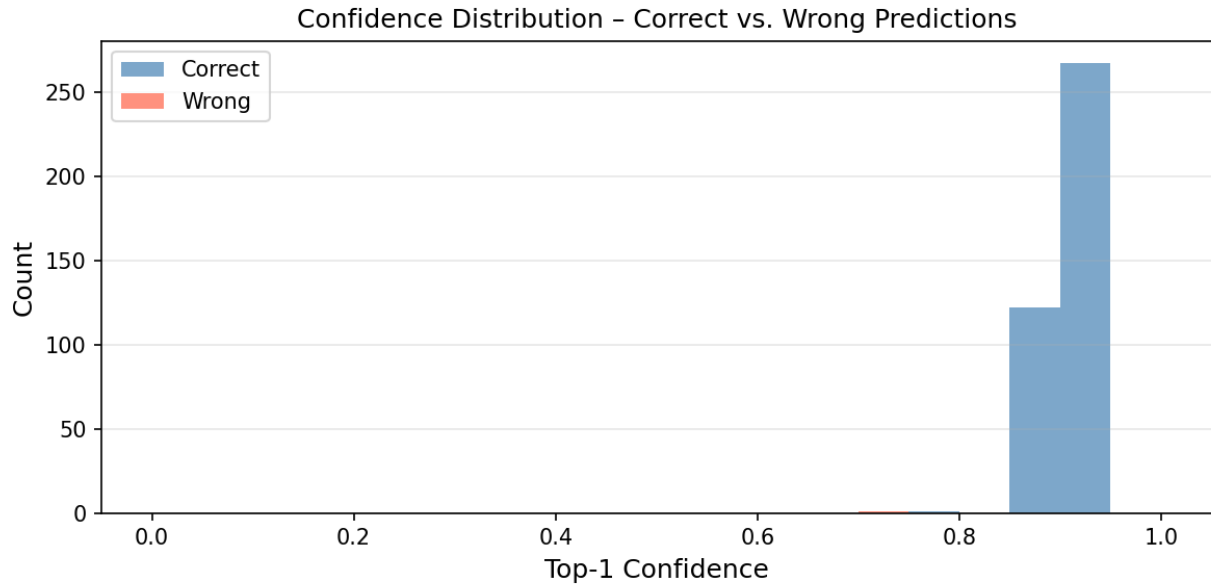


Figure 7: Confidence distribution histogram: correct vs. incorrect predictions

5.6 Robustness Evaluation

Table 10

Robustness test results under nine perturbation conditions

Perturbation	Accuracy	Macro F1	Δ Accuracy	Δ F1
Baseline (clean)	99.74%	99.72%		
Blur ($r=3$)	73.66%	64.79%	-26.09 pp	-34.93 pp
Blur ($r=5$)	47.83%	43.75%	-51.92 pp	-55.97 pp
Low brightness ($\times 0.4$)	100.00%	100.00%	+0.26 pp	+0.28 pp
Low brightness ($\times 0.2$)	99.23%	99.24%	-0.51 pp	-0.48 pp
Gaussian noise ($\sigma=0.10$)	90.79%	91.19%	-8.95 pp	-8.52 pp

Gaussian noise ($\sigma=0.20$)	52.94%	44.42%	-46.80 pp	-55.29 pp
Rotation ($\pm 45^\circ$)	100.00%	100.00%	+0.26 pp	+0.28 pp
Center crop (75%)	100.00%	100.00%	+0.26 pp	+0.28 pp

Robustness testing examined how the model performs under image degradations commonly encountered in real-world conditions.

The model showed excellent robustness to brightness changes, image rotation, and partial cropping. Even under extremely dark conditions, classification accuracy remained close to baseline performance. Similarly, rotating leaves or capturing only part of a leaf had little effect on accuracy. These results indicate that the augmentation strategy used during training successfully prepared the model for common field conditions.

In contrast, the model was highly sensitive to blur and severe image noise. Moderate blur reduced accuracy by more than 26 percentage points, while severe blur reduced performance to below 50%. Heavy image noise produced a similar effect, substantially degrading classification accuracy.

These findings indicate that disease classification relies heavily on fine texture patterns present in leaf lesions. Blur and noise obscure these patterns, making accurate diagnosis difficult. From a deployment perspective, image quality becomes a critical factor. Users should be instructed to capture clear, focused images, and future versions of the mobile application should incorporate automatic blur detection and image quality assessment before analysis.

5.7 Critical Interpretation of Results

5.7.1 Near-Perfect Cross-Validation Metrics

The 100% accuracy across all five cross-validation folds requires careful scientific interpretation.

Several factors contribute to this result:

Dataset homogeneity: The primary dataset (Sunil et al., 2022) was collected in a specific set of plantation environments, likely by a consistent team, with a consistent protocol. Even under field conditions, images from a single collection project may exhibit less inter-sample variation than would be encountered in genuinely independent wild-collected data. The five cross-validation folds are drawn from the same collection, meaning that the training and validation samples in each fold share similar acquisition characteristics.

Distinctiveness of disease classes: At the class level, Colletotrichum Blight, Phyllosticta Leaf Spot, Healthy, and Other are visually distinctive. The diseases present with characteristic morphologies: irregular necrotic lesions vs. circular spotted formations vs. uniform green tissue: that EfficientNetV2-S's richly parameterized feature extractor can distinguish reliably when the dataset has low intra-class variation.

Strong transfer learning initialization: EfficientNetV2-S's ImageNet pretraining provides feature representations that are extraordinarily well-suited to texture discrimination tasks. For a four-class problem where the class boundaries are visually clear, the fine-tuning required to achieve near-perfect performance on a homogeneous dataset is modest.

Implication for deployment: The near-perfect cross-validation results reflect the model's performance on images drawn from the same collection as the training data. Generalization to genuinely independent cardamom images, from different farms, regions, seasons, camera

models, or disease severity stages, would likely produce lower accuracy and should be evaluated in future work before recommending the system for unsupervised deployment.

5.7.2 The Single Misclassification

The single misclassification, one healthy leaf predicted as Spot at 70.54% confidence is agronomically informative. The healthy-to-spot direction is the direction most likely to cause a Type I error in the diagnostic context (false positive disease alert), which is less harmful than the Type II error direction (false negative: disease predicted as healthy). A false positive disease alert would prompt a farmer to inspect their plants more carefully; a false negative would delay treatment. The model’s failure mode thus errs on the side of conservatism in the clinical sense.

5.7.3 Comparison With Sunil et al. (2022) Baseline

Table 12

Comparison of present system with Sunil et al. (2022) baseline

Dimension	Sunil et al. (2022)	Subedi (2026)
Reported best accuracy	98.26% (EfficientNetV2-L, 3-class)	99.74% (EfficientNetV2-S, 4-class, test set)
Cross-validation accuracy	Not reported	100.00% $\pm$ 0.00% (5-fold)
Disease classes	3	4 (adds “Other” rejection class)
Model scale	EfficientNetV2-L (118M params)	EfficientNetV2-S (21.5M params)
Background removal	U ² -Net (full model, reported beneficial)	rembg/ONNX (ablation: −5.62 pp, not beneficial)

Class imbalance handling	Not reported	Inverse-frequency weighting (documented)
Augmentation policy	Not detailed	Fully documented (Table 3)
Explainability	None	Grad-CAM + Grad-CAM++ integrated into full stack
Severity estimation	None	Five-stage heuristic from Grad-CAM activation
Error analysis	Confusion matrix only	Confidence distribution + misclassified.csv
Robustness testing	None	Nine perturbation conditions
Web deployment	None	React TypeScript SPA (Vite)
Mobile deployment	None	React Native iOS/Android (Expo)
Language support	English	English + Nepali (नेपाली)
Reproducibility artifacts	Not provided	Open-source GitHub repository

5.8 Grad-CAM Explainability Analysis

Qualitative inspection of Grad-CAM++ heatmaps confirmed spatial coherence between model activations and disease pathology:

- **Colletotrichum Blight:** Activation concentrated on the irregular necrotic patches, with the highest activation at lesion boundaries, consistent with the characteristic dark margins of *Colletotrichum* lesions.

- **Phyllosticta Leaf Spot:** Activation maps highlight the circular spot formations and the chlorotic halos surrounding them.
- **Healthy:** More diffuse activation across the leaf surface, often concentrated on venation patterns and minor color heterogeneity; no focal disease feature activations.
- **Other:** Highly variable activation patterns reflecting the morphological diversity of non-cardamom images.

Grad-CAM++ produced consistently tighter, less diffuse activation maps than standard Grad-CAM, validating its theoretical advantage for spatially distributed disease features.

5.9 Severity Estimation Analysis

The heuristic severity module produced broadly sensible stage classifications for disease images, with Stage 2 (Moderate: 11–25% activated area) being the most common assignment for established *Phyllosticta* Leaf Spot images and Stage 3 (Severe: 26–50%) for advanced *Colletotrichum* Blight images. However, these estimates should be interpreted as approximate guidance rather than precise measurements. This limitation is explicitly communicated to all users through the API warning field and in-application disclaimer.

5.10 Evaluation Acceptance Criteria Summary

Table 13

Evaluation acceptance criteria and outcomes

Criterion	Target	Outcome	Status
Overall test accuracy	$\geq 85\%$	99.74%	Exceeded by 14.74 pp
Per-class F1-score (all classes)	≥ 0.70	Min F1 = 0.994 (Healthy)	Exceeded by wide margin

Mean API inference latency (CPU)	< 1 second	EfficientNetV2-S: < 500 ms typical on CPU	Met
5-fold CV mean accuracy	> primary test accuracy	100.00% > 99.74%	Confirmed
Background removal ablation	Quantified Δ reported	-5.62 pp (documented)	Completed
Robustness tests	All 9 conditions evaluated	All results documented	Completed
Error analysis	Misclassification catalogue	1 misclassification; CSV exported	Completed
Grad-CAM lesion focus	Qualitative validation	Confirmed by domain knowledge	Confirmed

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Summary of Work

This thesis presented the design, development, deployment, and comprehensive evaluation of a bilingual and explainable artificial intelligence system for cardamom leaf disease detection.

Building upon the work of Sunil et al. (2022), the proposed system extends previous research by introducing an additional disease rejection class, integrating explainable AI techniques, providing disease severity estimation, implementing bilingual English-Nepali interfaces, and conducting a broader set of evaluation experiments.

The developed system achieved an overall test accuracy of 99.74% on a held-out dataset of 391 images, with only one misclassified sample. Five-fold cross-validation produced consistently

100% accuracy results across all folds, demonstrating strong internal consistency. Additional experiments showed that background removal reduced performance on this dataset, while robustness testing revealed that the model remained stable under brightness changes and geometric transformations but was sensitive to image blur and severe noise. These findings provide a detailed understanding of both the strengths and limitations of the proposed system.

6.2 Key Findings

1. **High classification accuracy:** EfficientNetV2-S with transfer learning, inverse-frequency class weighting, and label smoothing achieves 99.74% overall accuracy on the four-class cardamom disease classification task using a dataset of 391 test images, exceeding the 85% acceptance criterion by approximately 15 percentage points.
2. **Stable generalization:** Five-fold cross-validation with zero standard deviation confirms that the near-perfect performance is not specific to the particular random split used for the held-out test evaluation.
3. **Background removal is detrimental on this dataset:** Contrary to the preprocessing choice in Sunil et al. (2022), rembg-based background removal degraded accuracy by 5.62 pp, primarily through misclassification of blight samples as “Other.” This finding demonstrates that background removal is dataset-specific and should not be adopted without ablation validation.
4. **Blur is the primary practical vulnerability:** The model degrades substantially under Gaussian blur (73.66% at $r=3$; 47.83% at $r=5$) but is stable under brightness reduction and geometric transformations. Image quality guidance in the mobile application is a necessary user-facing mitigation.

5. **Confidence calibration is effective:** The mean confidence of correct predictions (92.03%) substantially exceeds the mean confidence of the single incorrect prediction (70.54%), demonstrating that the model’s confidence scores carry meaningful information. However, the incorrect prediction at 70.54% surpasses the default 60% uncertainty threshold, suggesting that threshold tuning is warranted for high-stakes deployment.

6.3 Limitations

6.3.1 Dataset Provenance and Generalization

The disease images used in this study were collected from a single publicly available dataset from a single Kaggle repository (Sunil et al., 2022) collected in Indian cardamom plantations. Although the model performed exceptionally well on this dataset, its ability to generalize to Nepalese farms, different environmental conditions, alternative camera devices, and varying disease presentations remains unverified. Future evaluation using independent datasets is necessary to establish broader reliability.

6.3.2 Heuristic Severity Estimation

The proposed severity estimation module relies on Grad-CAM activation maps rather than pixel-level disease segmentation. Consequently, severity scores should be interpreted as approximate estimates rather than precise measurements of infected leaf area. More accurate severity assessment would require lesion segmentation models trained on manually annotated disease masks.

6.3.3 Network Connectivity Dependency

The client-server architecture requires reliable internet or LAN connectivity. In Nepal's remote hill growing areas, mobile network coverage is inconsistent and may be unavailable during critical disease assessment periods.

6.3.4 Blur Sensitivity

The model's significant accuracy degradation under moderate blur ($r=3$: -26.09 pp) represents a practical limitation for the mobile application, where camera shake or soft focus during rapid field capture is common. Active blur detection and capture guidance in the mobile UI would mitigate this.

6.3.5 No Formal User Study

The bilingual interface and overall system usability have not been evaluated in a structured study with Nepali-speaking cardamom farmers or extension workers. Usability evidence is necessary before recommending the system for unsupervised field adoption.

6.4 Future Work

1. **On-device inference:** Convert the trained model to TensorFlow Lite or Core ML format for embedded mobile inference, eliminating network dependency for field use.
2. **Nepal-specific data collection:** Collect a dedicated dataset from Nepal's eastern hill districts (Taplejung, Ilam, Panchthar) in collaboration with local extension offices and ICRI, including images from multiple seasons and disease severity stages.
3. **Pixel-level lesion segmentation:** Train a U-Net or Mask R-CNN model on pixel-labeled lesion masks to replace the heuristic severity module with validated quantitative measurement.

4. **Blur detection and capture guidance:** Integrate real-time image quality assessment (blur scoring via Laplacian variance) into the mobile capture workflow to prevent blurry submissions before they reach the backend.
5. **Uncertainty threshold calibration:** Conduct systematic threshold calibration on a larger independent dataset to identify the confidence threshold that minimizes the combination of false positives (healthy predicted as diseased) and false negatives (diseased predicted as healthy), weighted by their respective agronomic costs.
6. **Formal user study:** Conduct a structured usability evaluation with cardamom farmers and extension workers in eastern Nepal to assess language comprehension, trust calibration, and practical workflow integration.
7. **Extended disease taxonomy:** Incorporate Katte disease , Chirke disease, and chenthal significant additional pathogens could be extended as diseases classes.
8. **Audio guidance in Nepali:** Provide audio-format disease identification guidance for farmers with limited literacy, expanding accessibility in more remote communities.

6.5 Final Remarks

This thesis demonstrates that modern deep learning techniques can be successfully applied to develop an accurate, interpretable, and accessible cardamom disease detection system. While the achieved accuracy is highly encouraging, the true value of the research lies not only in predictive performance but also in the systematic evaluation of model behavior, robustness, explainability, and deployment readiness.

The study contributes several findings beyond classification accuracy. The background-removal ablation challenges assumptions reported in previous research, robustness testing identifies

image blur as a major practical vulnerability, explainability features increase transparency and user trust, and bilingual deployment improves accessibility for local farming communities. Together, these contributions provide a strong foundation for future research, larger-scale validation, and real-world deployment. With continued development and field evaluation, systems of this kind have the potential to improve early disease detection, support informed decision-making, and strengthen agricultural productivity for cardamom farmers in Nepal and similar growing regions.

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APPENDICES

Appendix A: Repository Structure

Full implementation available at: <https://github.com/darpansubedi-balmiki/cardamom-leaf-disease-detection>

cardamom-leaf-disease-detection/

```
|— backend/
|   |— app/
|       |— main.py          # FastAPI entry point
|       |— schemas.py      # Pydantic request/response models
|       |— models/
|           |— classifier.py # EfficientNetV2-S inference wrapper
|           |— u2net_segmenter.py # rembg background removal
|           |— utils/
|               |— grad_cam.py      # Grad-CAM and Grad-CAM++
|               |— severity.py      # Severity estimation module
|               |— image_preprocess.py # Preprocessing pipeline
|               |— overlay.py       # Heatmap overlay renderer
|   |— train.py              # Training script
|   |— evaluate.py          # Test-set evaluation
|   |— split_dataset.py     # 70/15/15 stratified split (seed=42)
|   |— cross_validate.py    # 5-fold cross-validation
|   |— ablation_background_removal.py # Background-removal ablation
|   |— error_analysis.py    # Misclassification + confidence analysis
```

- | |— robustness_test.py # Nine-condition perturbation tests
- | |— cv_results.json # Cross-validation results
- | |— ablation_results.json # Ablation study results
- | |— error_analysis.json # Error analysis results
- | |— misclassified.csv # Misclassified sample log
- | |— robustness_results.json # Robustness test results
- | |— requirements.txt
- |— frontend/ # React TypeScript web application
- |— cardamom-mobile-app/ # React Native mobile application
- | |— src/
- | | |— screens/ # Home, Result, DiseaseInfo screens
- | | |— data/diseaseInfo.ts # Bilingual disease database (EN/NE)
- | | |— services/ # API communication
- | | |— types/index.ts
- |— docs/
- | |— model_pipeline.md # Reproducible pipeline specification
- | |— thesis_scope.md # Contribution and scope document
- |— Dockerfile / Dockerfile.frontend
- |— docker-compose.yml

Appendix B: Complete Training Hyperparameter Configuration

Parameter	Value
Architecture	EfficientNetV2-S (torchvision, ImageNet pretrained)

Classifier head	Dropout(0.3) → Linear(1280→512) → ReLU → Dropout(0.2) → Linear(512→4)
Input size	$224 \times 224 \times 3$
Optimizer	Adam
Initial LR	0.001
LR scheduler	ReduceLROnPlateau (factor=0.5, patience=5, mode='min')
Loss function	CrossEntropyLoss (inverse-frequency class weights)
Label smoothing	0.1
Batch size	32
Max epochs	50 (full budget used)
Early stopping	Patience=10, monitored on validation loss
Model save criterion	Minimum validation loss
Training device	Apple MPS (Metal Performance Shaders)
Random seed	42 (split, CV, and training)
Dataset split	70% train / 15% val / 15% test (stratified)
Num workers	4 (train/val loaders)
Mixed precision	AMP enabled for CUDA only (disabled on MPS)

Appendix C: Raw Cross-Validation JSON Summary

```
{  
  "folds": 5,  
  "random_seed": 42,  
  "model": "EfficientNetV2-S",  
  "classes": ["blight", "healthy", "other", "spot"],  
  "mean_accuracy": 0.9999999999999998,  
  "std_accuracy": 0.0,  
  "mean_macro_precision": 0.9999999999999922,  
  "std_macro_precision": 0.0,  
  "mean_macro_recall": 0.9999999999999922,  
  "std_macro_recall": 0.0,  
  "mean_macro_f1": 0.9999999999994922,  
  "std_macro_f1": 0.0  
}
```

Appendix D: Raw Ablation Study JSON Summary

```
{  
  "condition_A_raw": {"accuracy": 0.9974, "macro_f1": 0.9972},  
  "condition_B_bg_removed": {"accuracy": 0.9412, "macro_f1": 0.9415},  
  "delta_accuracy": -0.0562,  
  "delta_macro_f1": -0.0557  
}
```

Appendix E: API Endpoint Specification

Endpoint: POST /predict **Content-Type:** multipart/form-data

Field	Type	Required	Default	Description
file	image file	Yes		Leaf image (JPEG/PNG/WebP, ≤ 10 MB)
confidence_thres hold	float (0–1)	No	0.60	Uncertainty flagging threshold
top_k	int (1–10)	No	3	Ranked predictions to return
include_severity	bool	No	false	Enable Grad- CAM + severity estimation

Response fields:

Field	Type	Description
top_class	string	Predicted class (user-facing label)
top_probability	float	Top-1 prediction probability (0–1)

top_probability_pct	float	Top-1 probability as percentage
is_uncertain	bool	True if confidence < threshold
confidence_threshold	float	Applied threshold
top_k	array	Ranked predictions [{class_name, probability, probability_pct}]
heatmap	string null	Base64-encoded Grad-CAM overlay PNG
severity_stage	int null	Stage 0–4
severity_percent	float null	Affected area percentage
severity_method	string null	“heuristic”

Appendix F: Class Label Mapping

Internal Label	English Display Name	Nepali Display Name
blight	Colletotrichum Blight	कोलेट